MODEL LAB · 2026-10-19

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	24,115	3.04%	0.6080
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-06, scored on the 24,115 that came after (actual HR rate 3.04%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5898	1.67×	0.02937	0.13496	0.00609
KNN comps Prior is the pooled rate of the nearest comparables.	0.6080	1.56×	0.02932	0.13372	0.00385
SVR Support-vector regression on the same feature matrix.	0.6031	1.58×	0.02934	0.13424	0.00490
Random forest Forest regression; a learned similarity metric.	0.6048	1.58×	0.02932	0.13379	0.00336
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6080	1.56×	0.02932	0.13372	0.00385
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6080	1.56×	0.02932	0.13372	0.00385
Linear regression Ordinary least squares on the same features.	0.6065	1.61×	0.02934	0.13395	0.00406
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6059	1.65×	0.02936	0.13406	0.00525
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6051	1.63×	0.02933	0.13387	0.00397
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6041	1.58×	0.02933	0.13393	0.00438

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **258** knn **258** svr **258** rf **258** core **258** ensemble **258** linear **258** logistic **258** nn **258** xgb **258**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,467 PA, actual HR rate 2.39%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5828	1.52×	0.02336	0.11312	0.01660
KNN comps Prior is the pooled rate of the nearest comparables.	0.6179	2.20×	0.02327	0.11072	0.01033
SVR Support-vector regression on the same feature matrix.	0.5974	1.35×	0.02342	0.11447	0.01757
Random forest Forest regression; a learned similarity metric.	0.6043	1.69×	0.02335	0.11144	0.01396
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6179	2.20×	0.02327	0.11072	0.01033
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6179	2.20×	0.02327	0.11072	0.01033
Linear regression Ordinary least squares on the same features.	0.6177	1.69×	0.02349	0.11254	0.01413
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6177	1.35×	0.02353	0.11259	0.01721
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6138	1.86×	0.02330	0.11111	0.01206
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6084	1.52×	0.02339	0.11150	0.01594

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Junior Caminero 17.1%, Jo Adell 16.0%, Jonathan Aranda 14.2%, Patrick Bailey 13.1%, Yandy Díaz 12.8%, Victor Mesa Jr. 12.8%
Flat league	6/6	Junior Caminero 16.4%, Jo Adell 15.2%, Jonathan Aranda 13.5%, Yandy Díaz 13.1%, Victor Mesa Jr. 12.6%, Patrick Bailey 12.5%
KNN comps	6/6	Junior Caminero 16.9%, Jo Adell 15.8%, Jonathan Aranda 14.6%, Patrick Bailey 13.0%, Victor Mesa Jr. 12.8%, Yandy Díaz 12.8%
SVR	6/6	Junior Caminero 18.3%, Jo Adell 16.3%, Jonathan Aranda 14.1%, Victor Mesa Jr. 13.4%, Yandy Díaz 13.4%, Patrick Bailey 13.0%
Random forest	6/6	Junior Caminero 17.4%, Jo Adell 15.9%, Jonathan Aranda 14.1%, Victor Mesa Jr. 13.4%, Yandy Díaz 13.2%, Patrick Bailey 12.8%
Core (KNN+SVR)	6/6	Junior Caminero 16.9%, Jo Adell 15.8%, Jonathan Aranda 14.6%, Patrick Bailey 13.0%, Victor Mesa Jr. 12.8%, Yandy Díaz 12.8%
Forest blend	6/6	Junior Caminero 16.9%, Jo Adell 15.8%, Jonathan Aranda 14.6%, Patrick Bailey 13.0%, Victor Mesa Jr. 12.8%, Yandy Díaz 12.8%
Linear regression	6/6	Junior Caminero 17.2%, Jo Adell 16.0%, Jonathan Aranda 14.2%, Victor Mesa Jr. 13.1%, Yandy Díaz 13.0%, Patrick Bailey 12.9%
Logistic regression	6/6	Junior Caminero 17.6%, Jo Adell 16.4%, Jonathan Aranda 14.0%, Victor Mesa Jr. 13.1%, Yandy Díaz 12.9%, Patrick Bailey 12.5%
Neural net (torch)	6/6	Junior Caminero 17.6%, Jo Adell 16.3%, Jonathan Aranda 14.1%, Victor Mesa Jr. 13.2%, Yandy Díaz 12.8%, Patrick Bailey 12.8%
XGBoost	6/6	Junior Caminero 17.5%, Jo Adell 16.0%, Jonathan Aranda 14.2%, Yandy Díaz 13.4%, Victor Mesa Jr. 13.0%, Patrick Bailey 12.9%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 **Junior Caminero** TB slate	**17.1%**	16.4%	16.9%	18.3%	17.4%	16.9%	16.9%	17.2%	17.6%	
2 **Jo Adell** CLE slate	**16.0%**	15.2%	15.8%	16.3%	15.9%	15.8%	15.8%	16.0%	16.4%	
3 **Jonathan Aranda** TB	**14.2%**	13.5%	14.6%	14.1%	14.1%	14.6%	14.6%	14.2%	14.0%	
4 **Patrick Bailey** CLE	**13.1%**	12.5%	13.0%	13.0%	12.8%	13.0%	13.0%	12.9%	12.5%	
5 **Yandy Díaz** TB	**12.8%**	13.1%	12.8%	13.4%	13.2%	12.8%	12.8%	13.0%	12.9%	
6 **Victor Mesa Jr.** TB	**12.8%**	12.6%	12.8%	13.4%	13.4%	12.8%	12.8%	13.1%	13.1%	
7 **Chase DeLauter** CLE	**11.9%**	12.0%	11.8%	11.9%	11.8%	11.8%	11.8%	11.8%	11.7%	
8 **Travis Bazzana** CLE	**11.7%**	11.7%	11.5%	11.3%	11.5%	11.5%	11.5%	11.6%	11.3%	
9 **Nathaniel Lowe** CLE	**11.3%**	11.2%	11.5%	11.3%	11.5%	11.5%	11.5%	11.4%	11.3%	
10 **José Ramírez** CLE	**10.3%**	10.6%	10.4%	10.4%	10.5%	10.4%	10.4%	10.7%	10.9%	
11 **Petey Halpin** CLE	**9.4%**	10.3%	9.3%	9.7%	9.4%	9.3%	9.3%	10.4%	9.9%	
12 **Jonny DeLuca** TB	**8.5%**	9.0%	8.2%	8.1%	7.7%	8.2%	8.2%	8.2%	8.4%	
13 **Liam Hicks** TB	**7.6%**	8.7%	7.5%	8.3%	7.4%	7.5%	7.5%	8.0%	7.9%	
14 **Brayan Rocchio** CLE	**7.2%**	7.8%	7.1%	7.1%	6.8%	7.1%	7.1%	6.9%	6.9%	
15 **Richie Palacios** TB	**6.5%**	7.4%	6.6%	6.3%	5.8%	6.6%	6.6%	6.4%	6.4%	
16 **Steven Kwan** CLE	**5.2%**	8.6%	5.6%	3.6%	4.3%	5.6%	5.6%	3.6%	5.3%	
17 **Taylor Walls** TB	**4.4%**	6.0%	4.4%	3.6%	3.7%	4.4%	4.4%	4.6%	4.5%	
18 **Chandler Simpson** TB	**2.7%**	5.9%	3.4%	1.5%	1.6%	3.4%	3.4%	1.5%	3.1%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is a prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-19 · model 2026.10.01-stack · built 2026-10-19 10:43:07 · live data 2026-10-19 10:43:07

Built for research and entertainment. Probabilities are estimates, not guarantees; never stake what you cannot afford to lose. Data: MLB Stats API play-by-play, lineups and live scores; ESPN injuries and bios.